INCH-POUND

A-A-55524/2D
27 October 2022
SUPERSEDING
A-A-55524/2C
07 July 2017

COMMERCIAL ITEM DESCRIPTION

SHUNT, INSTRUMENT
(EXTERNAL, 50 MILLIVOLT, 170 THROUGH 600 AMPERES [A] LIGHTWEIGHT TYPE)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

The complete requirements for procuring the instrument shunts described herein shall consist of this document and the issue in effect of A-A-55524.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This COMMERCIAL ITEM DESCRIPTION (CID) uses a classification system which is included in the PIN as shown in the following example (see NOTES).

AA55524	/2	—	001
CID number	Applicable CID Specification Sheet (two digits)		Dash number (see table 1)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. The shunt supplied to this CID shall be as specified herein (see figure 1).

Configurations and dimensions. Configurations and dimensions shall be specified in figure 1.

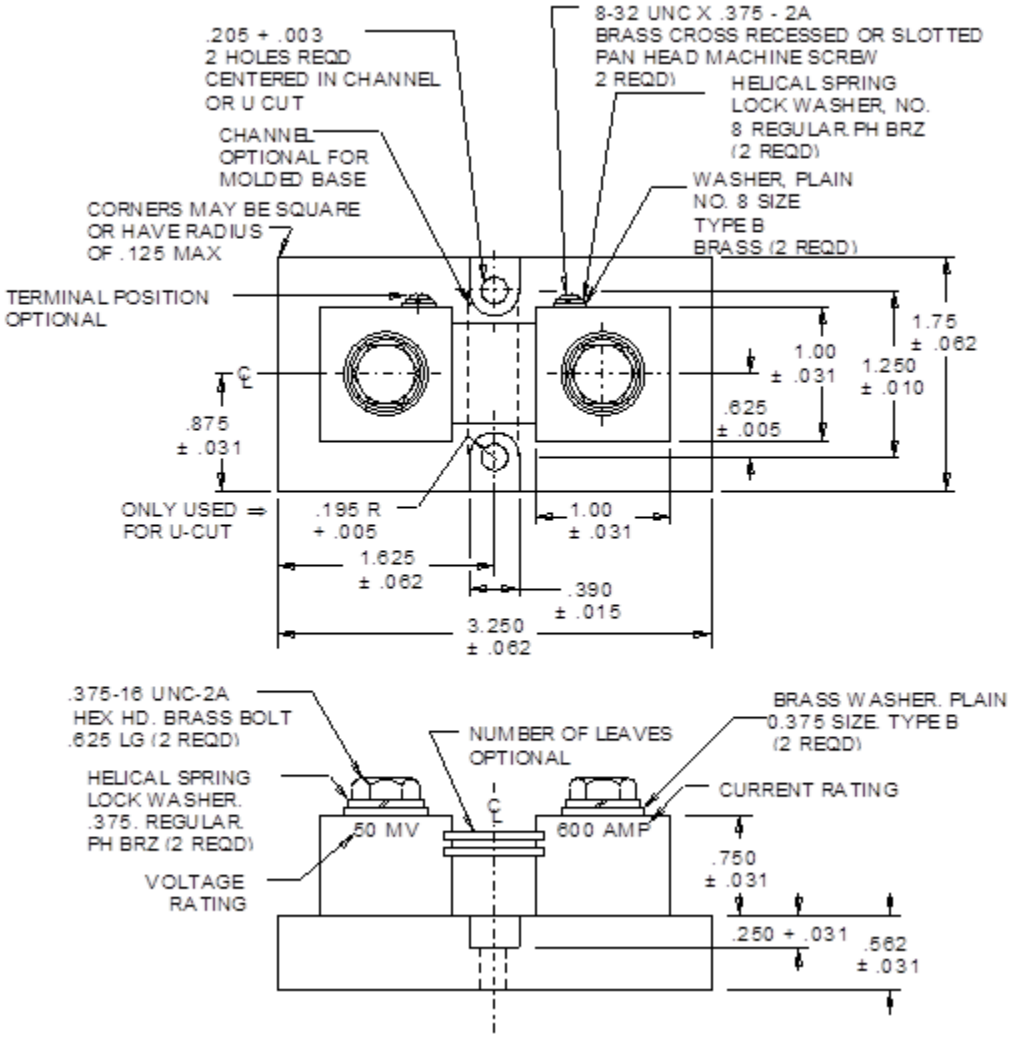
Mounting Hardware. Mounting hardware, screws and washers, shall be ANSI/ASME or equivalent.

Since Navy-EC is the Navy custodian for this document, all Navy review activities should forward their comments directly to DLA Land and Maritime. Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA-CC, DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990 or emailed to sound@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6625

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inches	mm	inches	mm
0.125	3.18	0.625	15.88
0.195	4.95	0.750	19.05
0.205	5.21	0.875	22.23
0.250	6.35	1.000	25.40
0.390	9.91	1.250	31.75
0.500	12.70	1.625	41.28
0.562	14.27	1.750	44.45
		3.250	82.55

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±0.005 inch (0.13mm).
3. The US government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

FIGURE 1. Configuration and dimensions.

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Ampere rating. The ampere rating shall be specified in table 1.

Voltage rating. The voltage rating shall be 50 mV ±0.3 mV.

Terminal block material. The terminal block material shall be brass.

Shunt resistance blade material. The resistance blade strip material shall be Manganin shunt metal or an alloy having an equivalent low temperature coefficient of resistance at any temperature, long term stability, and strength, within the working range of the shunt.

Shunt base mounting. The shunt-terminal blocks shall be mounted securely with screws of adequate size, and shall be made resistant to loosening. The heads of the screws shall be recessed in the base and properly protected with a good grade of insulating compound which shall not flow, crack, chip, pulverize, or otherwise deteriorate substantially.

Shunt base material. The shunt base material shall be molded or laminated plastic or equivalent.

Marking. Shunts supplied to this CID sheet shall be marked with the manufacturer's standard commercial PIN, mV rating, and current rating as specified on figure 1 and in tables I and II.

Style and type designation cross-reference. Table I is for cross-reference only. Ordering is identified in A-A-55524 (see note titled “Supersession data”).

Table I. Style and type designation cross-reference.

CID AA55524/2	Current Rating (Amperes)	MS91587	MIL-S-61 Type & Style	AN3200
-001	170	-1	MSB171	-170
-002	200	-2	MSB201	***
-003	250	-3	MSB251	-250
-004	300	-4	MSB301	-300
-005	400	-5	MSB401	-400
-006	450	-6	MSB451	-450
-007	500	-7	MSB501	***
-008	600	-8	MSB601	-600

QUALITY ASSURANCE PROVISIONS. Quality assurance provisions shall be as specified in A-A-55524.

PACKAGING. Packaging shall be as specified in A-A-55524.

Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order (see Marking in A-A-55524).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See the classification section, for PIN format example.

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Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these shunts to the DLA Land and Maritime under the Parts Management Advisory Team evaluation program, CAGE code 58536 should be used.

Source of documents.

Commercial Item Description

A-A-55524 - Shunts, Instrument (External, 50 Millivolt, Lightweight Type).

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Ordering data. Ordering data shall be as specified in A-A-55524.

Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer's CAGE	Manufacturer's name and address
03030	Empro Manufacturing Co., Inc. PO. Box 26060 10920 East 59th Street Indianapolis, IN 46226 Phone Number: 317-823-4478
0DRX0	DELTEC 13065 Tom White Way, Unit G Norwalk, CA 90650-8936 Phone Number: 562-926-2304, (Spectrum, 978-532-4400)
K6197	TYCO Electronics UK LTD Freebournes Road Witham, Essex CM8 3AH Phone Number: 01376 509509 Fax Phone Number: 01376 509512 CONUS: TE Energy: 1-847-841-8233
3AB55	Canadian Shunt Industries LTD 50 Bullock Drive, Unit 1 Markham, Ontario L3P 3P2 Phone Number: 905-294-5274
91812	Esterline Mason 13955 Balboa Blvd Sylmar, CA 91342-1084 Phone Number: 818-361-3366
09179	ELECTRICAL RESEARCH CORPORATION 1903 Barrett DR Troy, MI 48084-5372 Tel: (248) 362-6135

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Manufacturer's part number (P/N) supersession data. These CID specification sheet PIN's, as shown in table II, are used only within the Government to buy, store, and distribute commercial products and supersede the following manufacturers' P/N's, as shown. This information is being provided, to assist in reducing proliferation of part numbers within the Government inventory system.

TABLE II. Part number (P/N) supersession data.

Superseding CID PIN (see table I) AA55524/2-	Superseded manufacturers' CAGE codes and P/N's 1/					
	03030	0DRX0	K6197	3AB55	91812	09179
001	MLB-170- 50	MKB-170- 50	TM-170-50	LB-170-50	8300-170	21M170A50
002	MLB-200- 50	MKB-200- 50	TM-200-50	LB-200-50	8300-200	21M200A50
003	MLB-250- 50	MKB-250- 50	TM-250-50	LB-250-50	8300-250	21M250A50
004	MLB-300- 50	MKB-300- 50	TM-300-50	LB-300-50	8300-300	21M300A50
005	MLB-400- 50	MKB-400- 50	TM-400-50	LB-400-50	8300-400	21M400A50
006	MLB-450- 50	MKB-450- 50	TM-450-50	LB-450-50	8300-450	21M450A50
007	MLB-500- 50	MKB-500- 50	TM-500-50	LB-500-50	8300-500	21M500A50
008	MLB-600- 50	MKB-600- 50	TM-600-50	LB-600-50	8300-600	21M600A50

1/ The manufacturers' P/N's shall be used for actual part marking and shall not be used during procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial product part numbers were reviewed and replaced by the CID specification sheet PIN shown. For actual part marking requirements, see figure 1, table I, and table II.

Government users. To acquire information on obtaining these shunts from the Government inventory system, contact DLA Land and Maritime, ATTN: FMX, Post Office Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-7790. Division Chief email: MaritimeSupplierOps.FMX@dla.mil, phone: DSN 850-3490 or COMM (614) 692-3490.

Standard requirements for testing. Recommended testing requirements are as follows.

Reference Test Conditions. Reference test conditions are as follows.
Reference Temperature: 25 °C (±2 °C)
Position, Bus-Type: Current terminals and resistance blades, vertical, allowing free air flow.
Position, Portable Type: Base flat on horizontal surface.
Connections, Bus-Type Shunts: Tightly bolted to current bus. Use all terminal slots and blade insertion' specified depth.
Connections, Portable Type Shunts: Tightly bolted to lugs with cables of proper cross-section.

Short Circuit Rating. Shunts shall be able to withstand ten times their rated current for one second without incurring any visible damage or loss in accuracy.

Adjustment Accuracy. Shunts shall be within a quarter percent of their nominal values. Applied current shall not raise the terminal blocks' temperature more than 10 °C above the ambient temperature.

Rated Accuracy. Shunts rated up to 1000 amperes shall have 0.33 percent tolerance. Shunts rated for at least 1000 amperes shall have 0.5 percent tolerance.

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External Temperature Influence. Raising the ambient temperature from 25 °C to 100 °C shall not cause more than a 0.25 percent change in accuracy for shunts rated up to 1000 amperes. No more than a 0.5 percent change shall occur for shunts rated at least 1000 amperes. Shunts shall be at 100 °C for four consecutive hours prior to measurement.

Temperature Rise. Shunts shall be exposed to 110 percent of their rated current for 24 hours. There shall be no physical damage or permanent accuracy change beyond the rated accuracy limits. This test shall be made in a bus-bar arrangement with buses extended at least three feet into the air on either side of the shunt.

Thermal EMF. Maintain a temperature difference of 15 °C between terminal blocks. The thermal electromotive force shall not exceed 0.1 percent of the rated voltage drop.

Effect of Shifting Current Terminal Connections on Multiple Terminal Shunts. Use half the normal number of connections. The connections used shall be in the two extreme positions, and copper blades shall be inserted into unused slots. There shall not be more than a one percent measured output voltage drop. The current used shall not raise the blocks' temperature over than 10°C above the ambient temperature.

Dielectric Test. Shunts with insulating bases shall be tested with a 60 Hz, 2600 volts rms current applied for one minute between the shunt terminals and a metallic platform on which the shunt is mounted.

Voltage Drop. When using the rated current, the shunt's standard voltage output at the instrument terminals shall be 50 or 100 millivolts.

Preferred Current Ratings (in Amps). Preferred current ratings are as follows.

1	5	10	15	25
50	75	100	150	200
250	300	400	500	600
750	800	1,000	1,200	1,500
2,000	2,500	3,000	4,000	5,000
6,000	8,000	10,000		

National Stock Numbers (NSN's). Table III is a list of NSN's assigned, which correspond to this CID specification sheet. The list is for information only and may not be indicative of all possible NSN's associated with the CID specification sheet. For up to date information on assigned NSN's, please contact DLA Land and Maritime, ATTN: FMX, Post Office Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-7790. Division Chief email: MaritimeSupplierOps.FMX@dla.mil, phone: DSN 850- 3490 or COMM (614) 692-3490.

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TABLE III. National stock numbers (NSN's).

Dash number (see table I) AA55524/2-	Current Rating (Amperes)	NSN
001	170	6625-01-322-1935 1/
002	200	6625-00-166-1013
003	250	6625-00-818-3909
004	300	6625-00-166-1005
005	400	6625-00-720-2854
006	450	6625-00-555-2254
007	500	6625-00-555-2256
008	600	6625-01-086-0760

1/ This NSN references obsolete MS91587-1 and there is no Current Procurement History Information found for NSN: 6625-01-322-1935.

Changes from previous issue. The margins of this CID specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - EC
Air Force - 71
DLA - CC

Review activities:
Army - AR, MI

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-7FXE

Preparing Activity:

DLA - CC

(Project 6625-2022-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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